Advanced Models

Problem

The objective of this project is to simulate steady-state airflow through a rectangular duct at an inlet Reynolds number of 1205, using different turbulence models. The turbulence models to be applied are: Zero Equation model, Spalart–Allmaras model, Shear Stress Transport (SST) model, and Baseline Explicit Algebraic Reynolds Stress Model (BSL EARSM).

The simulation results will be compared to Direct Numerical Simulation (DNS) data, focusing on the tangential velocity vectors at the duct outlet. Additionally, the convergence time of each turbulence model will be analyzed and compared, providing insight into the computational efficiency and stability of each approach.

Solution

The accurate prediction of turbulent flow behavior in ducts is essential for many engineering applications, such as ventilation systems, heat exchangers, and aerodynamic devices. Due to the complexity of turbulence, direct numerical simulation (DNS) is often used as a benchmark, but it is computationally expensive and impractical for most real-world problems. Therefore, various turbulence models have been developed to approximate the effects of turbulence with reasonable accuracy and computational cost.

This study aims to simulate steady airflow inside a rectangular duct using four different turbulence modeling approaches: the Zero Equation model, the Spalart–Allmaras model, the Shear Stress Transport (SST) model, and the Baseline Explicit Algebraic Reynolds Stress Model (BSL EARSM). The flow conditions are set at an inlet Reynolds number of 1205, with air at 25°C and 101.130 kPa, and a duct cross-sectional area of 4 cm².

The results from these turbulence models will be compared against DNS data, with a particular focus on the tangential velocity vectors at the duct outlet. Furthermore, the convergence time for each turbulence model will be evaluated and discussed to assess the trade-offs between computational efficiency and simulation accuracy.

Setup

The computational domain consists of a rectangular duct with a cross-sectional area of 4 cm². A structured mesh was generated to discretize the domain, ensuring sufficient refinement near the walls to capture boundary layer effects, presented on the Fig. 1. The simulations were performed under steady-state conditions. Appropriate solver settings were used for each turbulence model to ensure numerical stability, and convergence was monitored based on residual values, requiring at least a drop of 1×10^{-4} for all residuals.

The following boundary conditions were applied:

- **Inlet:** A uniform velocity profile was imposed, corresponding to a Reynolds number of 1205 based on the hydraulic diameter, which gives 0.94 m/s inlet velocity. Turbulence intensity and viscosity ratio were set according to standard inlet turbulence conditions, depending on the turbulence model. The fluid was modeled as incompressible air at a temperature of 25°C and an absolute pressure of 101.130 kPa.
- **Walls:** No-slip boundary conditions were imposed at all duct walls.
- **Symmetry:** The domain's lower boundary, along $y = 0$, was modeled as the axis of symmetry.
- **Outlet:** At the outlet, A fixed static pressure boundary condition was applied with gauge = absolute pressure of p = 101.130 kPa.

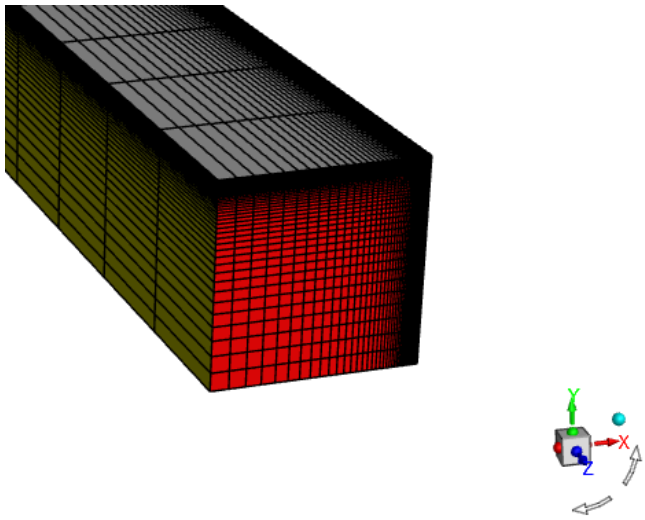


Figure 1: Computational domain and mesh for the rectangular duct in ANSYS Fluent.

Four turbulence models were employed in the simulations: Zero Equation model (Turbulent Mixing-length), Spalart–Allmaras model, Shear Stress Transport (SST) model and Baseline Explicit Algebraic Reynolds Stress Model (BSL EARSM). The simulations were initialized using a standard procedure, where the initial velocity and turbulence fields were specified according to the inlet conditions.

After convergence, post-processing was performed by capturing vector plots of the velocity magnitude at the outlet. Run time data for each simulation was also collected to evaluate the computational cost associated with each turbulence model.

Explicit Algebraic Reynolds Stress Models (EARSM) $k-\omega$

Explicit Algebraic Reynolds Stress Models (EARSM) represent an extension of the standard two-equation models. They are derived from the Reynolds stress transport equations and give a nonlinear relation between the Reynolds stresses and the mean strain-rate and vorticity tensors. Due to the higher order terms, many flow phenomena are included in the model without the need to solve transport equations for individual Reynolds stresses. The WJ-BSL-EARSM allows an extension of the baseline (BSL) turbulence model to capture the anisotropy of Reynolds stresses and secondary flows [1].

With EARSM, the Reynolds stresses are computed from the anisotropy tensor according to its definition:

$$\overline{u_i u_j} = k\left(a_{ij} + \frac{2}{3}\delta_{ij}\right), \tag{1}$$

where the anisotropy tensor a_{ij} is resolved as a solution of the following implicit algebraic matrix equation:

$$N\mathbf{a} = -A_1\mathbf{S} + (\mathbf{a}\Omega - \Omega\mathbf{a}) - A_2(\mathbf{aS} - \mathbf{Sa} - \frac{2}{3}tr\{\mathbf{aS}\}, \quad \text{with} \quad N = A_3 + A_4(P_k/\varepsilon). \tag{2}$$

The coefficients A_i in this matrix equation depend on the C_i-coefficients of the pressure-strain term in the underlying Reynolds stress transport model. Their values are selected here as A_1=1.245, A_2=0, A_3=1.8, A_4=2.25.

S_{ij} and Ω_{ij} denote the non-dimensional strain-rate and vorticity tensors, respectively. They are defined as:

$$S_{ij} = \frac{1}{2}\tau\left(\frac{\partial U_i}{\partial x_j} + \frac{\partial U_j}{\partial x_i}\right) \tag{3}$$

$$\Omega_{ij} = \frac{1}{2}\tau\left(\frac{\partial U_i}{\partial x_j} - \frac{\partial U_j}{\partial x_i}\right) \tag{4}$$

where the time scale τ is given by

$$\tau = k/\varepsilon = 1/(C_\mu \omega), \qquad C_\mu = 0.09. \tag{5}$$

Shear-Stress Transport (SST) $k-\omega$

The SST $k-\omega$ model accounts for the transport of the turbulence shear stress in the definition of the turbulent viscosity. These features make the SST model more accurate and reliable for a wider class of flows (for example, adverse pressure gradient flows, airfoils, transonic shock waves) than the above mentioned standard models [2]. Other models fail to properly predict the onset and amount of flow separation from smooth surfaces. The main reason is that other models do not account for the transport of the turbulent shear stress. This results in an overprediction of the eddy-viscosity. The proper transport behavior can be obtained by a limiter to the formulation of the eddy-viscosity:

$$\mu_t = \frac{\rho k}{\omega}\frac{1}{max[\frac{1}{\alpha^*}, \frac{SF_2}{a_1\omega}]}, \tag{6}$$

where S is the strain rate magnitude and α^* is defined as

$$\alpha^* = \alpha^*_\infty\left(\frac{\alpha^*_0 + Re_t/R_k}{1 + Re_t/R_k}\right), \quad Re_t = \frac{\rho k}{\mu\omega}, \quad R_k = 6, \quad \alpha^*_0 = \frac{0.072}{3}. \tag{7}$$

F_2 is given by

$$F_2 = \tanh\Phi_2^2, \qquad \Phi_2 = max\left[2\frac{\sqrt{k}}{0.09\omega y}, \frac{500\mu}{\rho y^2\omega}\right], \tag{8}$$

where y is the distance to the next surface.

One-Equation Spalart-Allmaras Model

The Spalart-Allmaras model [621] is a one-equation model that solves a modeled transport equation for the kinematic eddy (turbulent) viscosity. The Spalart-Allmaras model was designed specifically for aerospace applications involving wall-bounded flows and has been shown to give good results for boundary layers subjected to adverse pressure gradients. It is also gaining popularity in turbomachinery applications [3].

The transported variable in the Spalart-Allmaras model, $\widetilde{\nu}$, is identical to the turbulent kinematic viscosity except in the near-wall (viscosity-affected) region. The transport equation for the modified turbulent viscosity $\widetilde{\nu}$ is

$$\frac{\partial}{\partial t}(\rho\widetilde{\nu}) + \frac{\partial}{\partial x_i}(\rho\widetilde{\nu}u_i) = G_\nu + \frac{1}{\sigma_{\widetilde{\nu}}}\left[\frac{\partial}{\partial x_j}\left\{(\mu + \rho\widetilde{\nu})\frac{\partial\widetilde{\nu}}{\partial x_j}\right\} + C_{b2}\rho\left(\frac{\partial\widetilde{\nu}}{\partial x_j}\right)^2\right] - Y_\nu + S_{\widetilde{\nu}} \tag{9}$$

where G_ν is the production of turbulent viscosity, and Y_ν is the destruction of turbulent viscosity that occurs in the near-wall region due to wall blocking and viscous damping. $\sigma_{\widetilde{\nu}}$ and C_{b2} are the constants and ν is the molecular kinematic viscosity. $S_{\widetilde{\nu}}$ is a user-defined source term. Note that the turbulence kinetic energy, k, is not calculated in the Spalart-Allmaras model [3].

The turbulent viscosity μ_t is computed from

$$\mu_t = \rho \widetilde{\nu} f_{v1} \tag{10}$$

$$f_{v1} = \frac{\chi^3}{\chi^3 + C_{v1}}, \qquad C_{v1} = 0.71 \tag{11}$$

$$\chi \equiv \frac{\widetilde{\nu}}{\nu} \tag{12}$$

Mixing-Length Model

The mixing length model is a simple eddy viscosity model used for describing turbulence in shear flows. It assumes that turbulent eddies behave like molecules in a laminar fluid, transferring momentum over a characteristic length called the mixing length (l). Unlike modern two-equation models (e.g., $k-\varepsilon$, $k-\omega$), it does not solve transport equations for turbulence quantities but instead expresses eddy viscosity as a function of velocity gradients [4].

The turbulent eddy viscosity is given by

$$\mu_t = l^2 \left| \frac{dU}{dy} \right|, \tag{13}$$

where l is mixing length, U is the mean field velocity and y is the distance from the wall. This mode assumes that mixing length varies with position, typically using empirical formula:

$$l = \kappa y, \tag{14}$$

where $\kappa \approx 0.41$, and y is the distance from the wall.

Results & Discussion

The DNS result, Fig. 2, shows a complex secondary flow structure with small counter-rotating vortices in each corner of the duct cross-section, which is characteristic of low-Re turbulent duct flow.

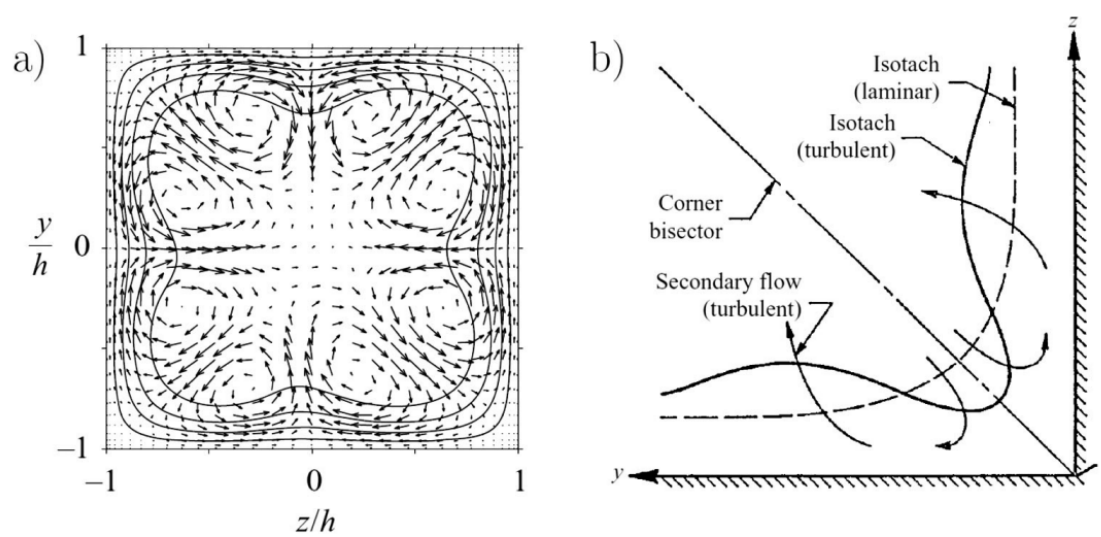
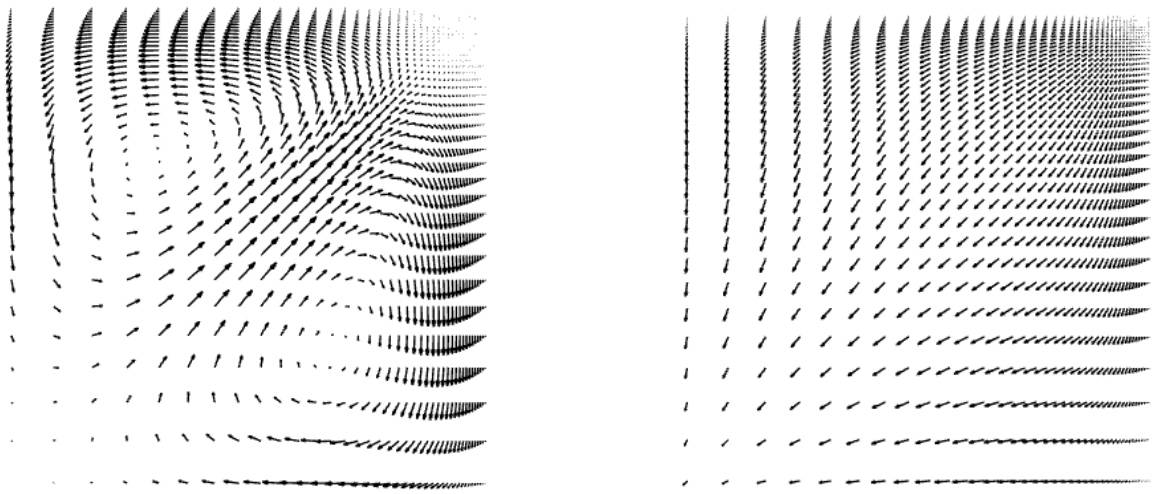


Figure 2: (a) Contour lines of the primary mean flow and vectors of the secondary mean flow for a DNS at a Reynolds number based on bulk velocity Re = 1205. (b) Sketch of the mean flow in the corner region of a straight duct. Taken from [5].

These corner vortices are a result of imbalance in Reynolds stresses and pressure gradients near the walls.

Among the RANS models tested, the BSL EARSM model was the only one that qualitatively captured secondary flows, Fig. 3a. However, it produced only two large vortices sweeping flow towards the upper right corner, failing to reproduce the detailed four-corner vortex structure seen in the DNS. This is likely due to the model's ability to resolve some anisotropy in the Reynolds stress tensor, but still being limited in its representation of the finer turbulence structures.



(a) Secondary flow vectors at the duct outlet predicted using the BSL EARSM model. The model captures two dominant counter-rotating vortices, indicative of partial resolution of secondary flows due to its anisotropy-aware formulation.

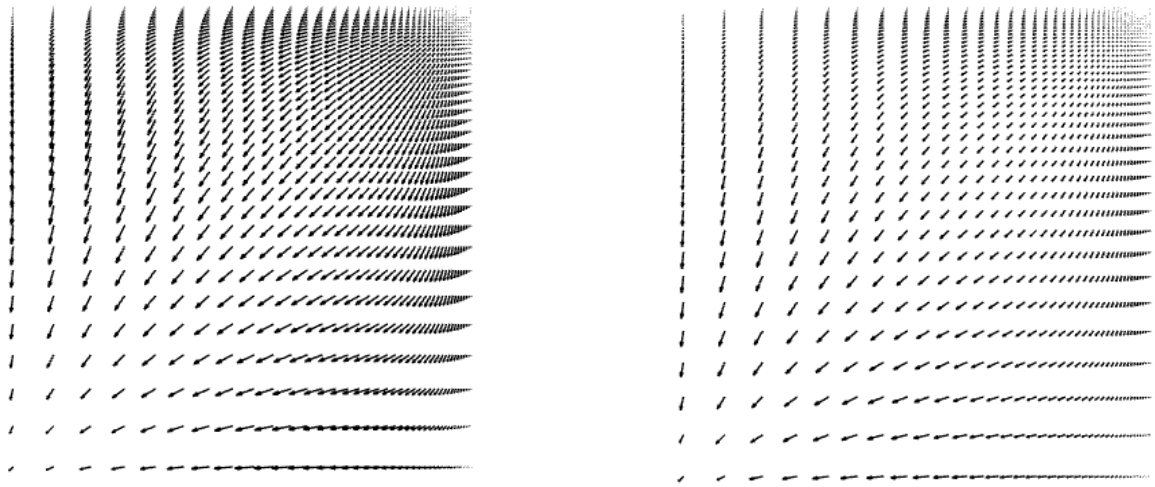
(b) Secondary flow vectors at the duct outlet predicted using the SST $k-\omega$ model. The flow pattern is largely unidirectional, with weak indications of secondary motion and poor representation of corner vortices.

Figure 3: Flow vectors at the duct outlet for BSL EARSM and SST $k-\omega$ models.

The other models — Spalart–Allmaras, SST $k-\omega$, and the Zero Equation (Mixing Length) model - all produced similar, oversimplified flow patterns, Fig. 3b, 4a and 4b. In these cases, velocity vectors showed a dominant single-directional motion towards the lower left corner, with little or no indication of secondary flow development in other corners. The fact that the vector scale had to be increased to visualize the motion suggests that these models underpredicted the transverse velocity components, likely due to their isotropic eddy-viscosity assumptions and lack of Reynolds stress anisotropy modeling.

Only the BSL EARSM model captured secondary flow structures, albeit in a simplified form with two dominant vortices, while the simpler eddy-viscosity models failed to resolve the corner vortices observed in the DNS and showed flow biased toward a single direction. The failure of the Spalart–Allmaras, SST, and mixing-length models to predict secondary flow arises from their reliance on isotropic eddy-viscosity assumptions. These models cannot capture turbulence-induced anisotropy or the cross-stream Reynolds stress gradients that drive secondary motion in ducts.

The failure of the Spalart–Allmaras, SST, and mixing-length models to predict secondary flow arises from their reliance on isotropic eddy-viscosity assumptions. These models cannot capture turbulence-induced anisotropy or the cross-stream Reynolds stress gradients that drive secondary motion in ducts. The need to manually increase the velocity vector scale in the SA, SST, and zero-equation cases further illustrates the models' inability to generate substantial secondary velocity compo-



(a) Secondary flow vectors at the duct outlet using the Spalart–Allmaras model. The flow shows predominantly one-directional behavior, with minimal transverse motion and no evidence of secondary vortices.

(b) Secondary flow vectors at the duct outlet using the Mixing Length (Zero-Equation) model. The solution shows a simplified flow structure lacking any secondary flow features or corner vortices.

Figure 4: Flow vectors at the duct outlet for one-equation Spalart-Allmaras and zero-equation mixing-length models.

nents, confirming that the predicted flows were predominantly unidirectional.

For engineering applications where secondary flows influence heat transfer or particle deposition (e.g., in cooling channels or clean rooms), relying on simple turbulence models may lead to inaccurate design predictions due to their inability to resolve these critical flow features.

Another trend observed in the outlet velocity vector plots is that the maximum velocity magnitude decreased progressively with more advanced turbulence models. For instance, the mixing-length model and Spalart–Allmaras showed higher peak velocities near the wall or lower left corner regions, while the SST model predicted slightly lower peak values. The BSL EARSM model yielded the lowest maximum vector magnitudes among all RANS simulations.

This behavior can be attributed to the increased physical realism and diffusivity of the more advanced models. Simpler models often overpredict turbulence mixing, leading to exaggerated secondary motion or spurious peak velocities. In contrast, models like SST and especially EARSM account for more accurate near-wall damping and turbulence anisotropy, leading to a more diffused and realistic velocity field with smoother gradients and lower peaks.

The reduced vector magnitudes in more advanced models may also reflect a suppression of unphysical flow features that simpler models are unable to filter out due to their limitations in handling complex 3D effects and low-Re behavior.

Convergence times

The convergence times for the four turbulence models vary significantly, as shown in Table 1. The Zero-Equation and Spalart–Allmaras models converged the fastest, requiring only 57 and 58 iterations respectively, with total simulation times of approximately 3 minutes. These models are relatively simple and involve solving fewer equations, with Spalart–Allmaras

being a one-equation model and the Zero-Equation model using algebraic closure without solving any additional PDEs.

Model	Average time per iteration [s]	Total time [s]	Number of iterations
BSL EARSM	2.956	387.201	131
SST	3.501	374.622	107
Spalart-Allmaras	2.873	166.611	58
Zero-Equation	3.218	185.417	57

Table 1: Comparison of convergence performance for the four turbulence models. The table shows the average time per iteration, total simulation time, and number of iterations required for each model to converge.

The SST $k-\omega$ model required 107 iterations, taking longer in terms of both iteration count and average time per iteration due to its two-equation structure and additional complexity near walls. The BSL EARSM model took the longest to converge (131 iterations, 6.5 minutes), despite a slightly faster per-iteration time than SST. This can be attributed to the model's increased mathematical complexity — particularly the algebraic Reynolds stress model — which adds stiffness to the solution and makes convergence more challenging. These results confirm the expected trade-off between model fidelity and computational cost: simpler models converge faster but lack physical detail, while more advanced models like BSL EARSM provide better accuracy at the expense of longer simulation times and more iterations.

Conclusion

This study compared the performance of four turbulence models — Zero-Equation (Mixing Length), Spalart–Allmaras, SST $k-\omega$, and BSL EARSM — in predicting the flow behavior in a square duct with emphasis on secondary flow structures and computational cost. The DNS results revealed complex secondary flow features driven by Reynolds stress anisotropy, especially small counter-rotating vortices in each corner of the duct. Among the RANS models, only BSL EARSM was able to partially reproduce these features, capturing two dominant vortices and reflecting some degree of anisotropy modeling. The simpler eddy-viscosity models (Spalart–Allmaras, SST, and Mixing Length) failed to resolve corner vortices and showed mostly unidirectional flow, underpredicting secondary motion and resulting in artificially high peak tangetial velocities in specific regions.

From a numerical standpoint, simpler models converged significantly faster, with the Zero-Equation and Spalart–Allmaras models completing in less than half the time required by SST and BSL EARSM. However, this came at the cost of physical accuracy, particularly in predicting 3D flow effects essential for engineering applications involving heat transfer or mixing. Overall, the results highlight the trade-off between computational efficiency and modeling fidelity. While simpler models may be suitable for bulk flow estimation in duct geometries, capturing key features like secondary flows and near-wall effects requires more advanced models such as BSL EARSM, despite their higher computational cost.

References

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[2] Inc. ANSYS. *SST k-omega Model*, 2025.

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[5] Ricardo Vinuesa, Azad Noorani, Adrián Lozano-Durán, George K El Khoury, Philipp Schlatter, Paul F Fischer, and Hassan M Nagib. Aspect ratio effects in turbulent duct flows studied through direct numerical simulation. *Journal of Turbulence*, 15(10):677–706, 2014.